应用电磁学基础 • 第8版

中英对照 • 章节总结复核修订版

在三轮复核修订版基础上，继续检查十个章节的总结与公式汇总页面。

本次修订8个总结页及1个正文页，恢复公式排版，补齐术语并校正矢量运算说明。

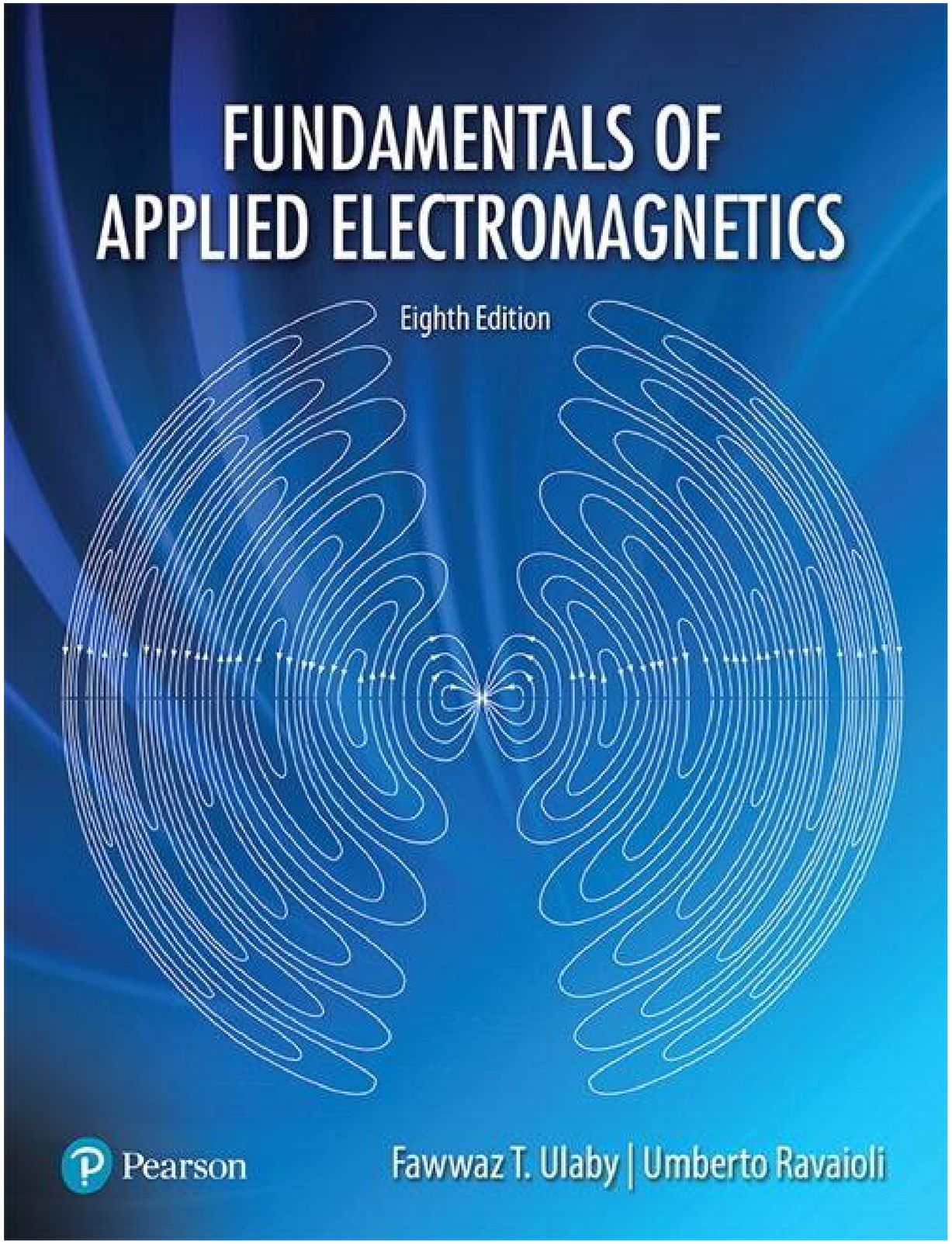
保留此前精校与局部修订；本次并非全书逐句精校或全部公式的独立推导。

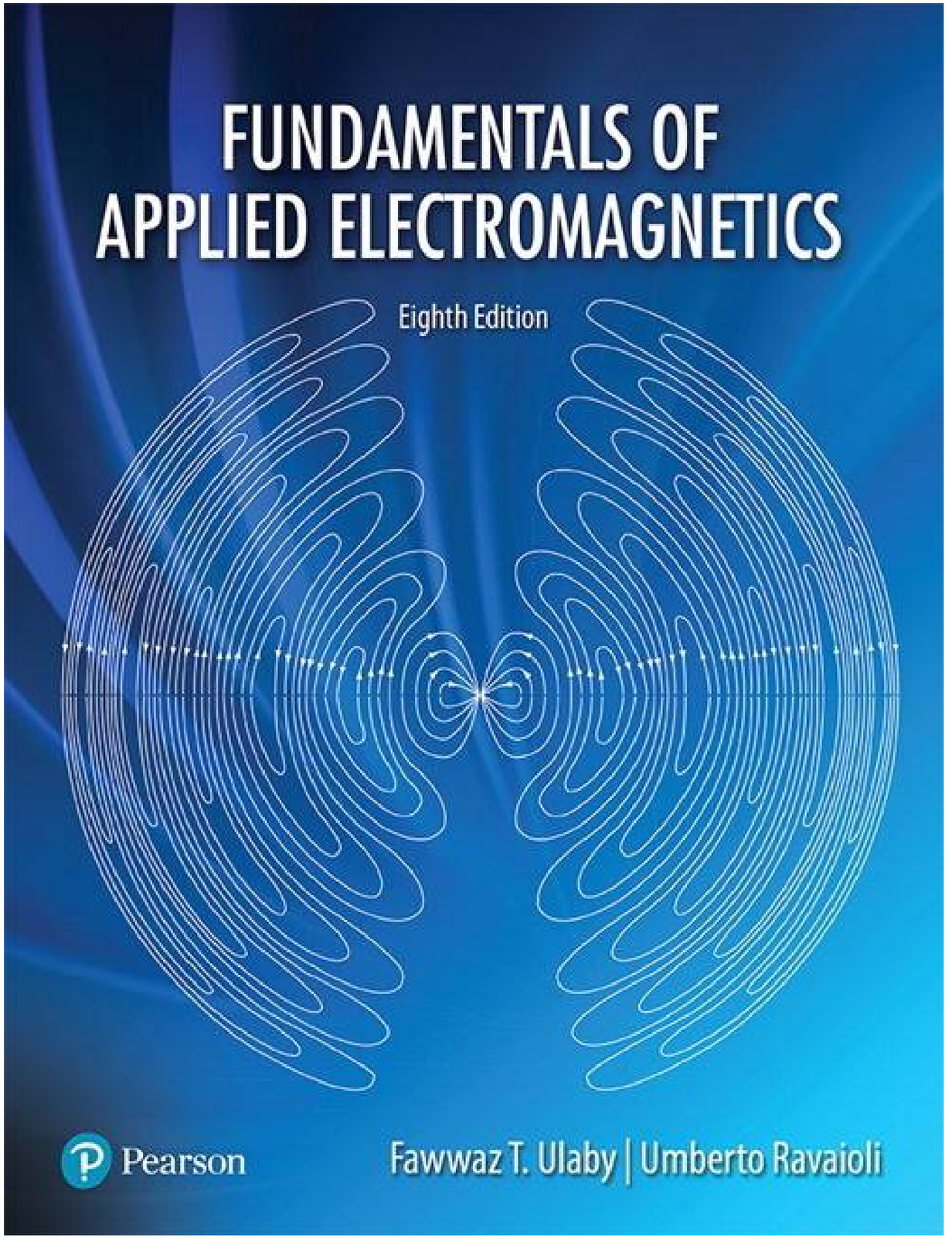
对易误用公式补充必要条件，并依据BIPM资料校正SI基本单位相关表述。

左侧保留英文原书。逐页修改项目及验证结果见同目录章节总结复核记录。

部分其他页面仍可能存在机翻问题，不能将本版本视为正式审定译本。

2026-09-13 | 源PDF页码与对照文件页码相差1页（新增封面）。


FUNDAMENTALS OF
APPLIED ELECTROMAGNETICS
Eighth Edition
Pearson
Fawwaz T. Ulaby | Umberto Ravaioli


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We dedicate this book to
Jean and Ann Lucia.

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Building on the core content and style of its predecessor, this eighth edition (8/e) of *Applied Electromagnetics* includes features designed to help students develop deep understanding of electromagnetic concepts and applications. Prominent among them is a set of 52 web-based simulation modules* that allow the user to interactively analyze and design transmission line circuits; generate spatial patterns of the electric and magnetic fields induced by charges and currents; visualize in 2-D and 3-D space how the gradient, divergence, and curl operate on spatial functions; observe the temporal and spatial waveforms of plane waves propagating in lossless and lossy media; calculate and display field distributions inside a rectangular waveguide; and generate radiation patterns for linear antennas and parabolic dishes. These are valuable learning tools; we encourage students to use them and urge instructors to incorporate them into their lecture materials and homework assignments.

Additionally, by enhancing the book's graphs and illustrations and by expanding the scope of topics of the Technology Briefs, additional bridges between electromagnetic fundamentals and their countless engineering and scientific applications are established.

NEW TO THIS EDITION

- Additional exercises
- Updated Technology Briefs
- Enhanced figures and images
- New/updated end-of-chapter problems

* The interactive modules and Technology Briefs can be found at the book companion website: **em8e.eecs.umich.edu**.

ACKNOWLEDGMENTS

As authors, we were blessed to have worked on this book with the best team of professionals: Richard Carnes, Leland Pierce, Janice Richards, Rose Kernan, and Paul Mailhot. We are exceedingly grateful for their superb support and unwavering dedication to the project.

We enjoyed working on this book. We hope you enjoy learning from it.

FAWWAZ T. ULABY
UMBERTO RAVAIOLI

第 8 版前言

本书《应用电磁学基础》第 8 版继承了前版的核心内容与风格，并加入多项有助于学生深入理解电磁学概念及应用的学习资源。其中尤为突出的是一套包含 52 个模块的网络仿真工具*。借助这些模块，读者可以交互式地分析和设计传输线电路；生成电荷和电流所激发的电场与磁场的空间分布；在二维和三维空间中观察梯度、散度和旋度对空间函数的作用；观察平面波在无损和有损介质中传播时的时间与空间波形；计算并显示矩形波导内的场分布；以及生成线天线和抛物面天线的辐射方向图。

这些模块是很有价值的学习工具。我们鼓励学生使用它们，也建议教师将其纳入课堂讲授和课后作业。

此外，本版改进了图表和插图，并扩展了“技术简介”专栏的选题范围，以进一步联系电磁学基础知识及其广泛的工程和科学应用。

本版新增内容

- 新增练习
- 更新“技术简介”专栏
- 改进图表和插图
- 新增或修订章末习题

* 交互模块和技术简介可在配套网站 em8e.eecs.umich.edu 获取。

致谢

在本书的编写过程中，我们有幸与一支优秀的专业团队合作，其成员包括 Richard Carnes、Leland Pierce、Janice Richards、Rose Kernan 和 Paul Mailhot。我们衷心感谢他们出色的支持，以及对本项目始终如一的投入。

我们很享受编写本书的过程，也希望你能从学习本书中获得乐趣。

Fawwaz T. Ulaby
Umberto Ravaioli

CONTENT

This book begins by building a bridge between what should be familiar to a third-year electrical engineering student and the electromagnetics (EM) material covered in the book. Prior to enrolling in an EM course, a typical student will have taken one or more courses in circuits. He or she should be familiar with circuit analysis, Ohm's law, Kirchhoff's current and voltage laws, and related topics.

Transmission lines constitute a *natural* bridge between electric circuits and electromagnetics. Without having to deal with vectors or fields, the student will use already familiar concepts to learn about wave motion, the reflection and transmission of power, phasors, impedance matching, and many of the properties of wave propagation in a guided structure. All of these newly learned concepts will prove invaluable later (in Chapters 7 through 9) and will facilitate the learning of how plane waves propagate in free space and in material media. Transmission lines are covered in Chapter 2, which is preceded in Chapter 1 with reviews of complex numbers and phasor analysis.

The next part of this book, contained in Chapters 3 through 5, covers vector analysis, electrostatics, and magnetostatics. The electrostatics chapter begins with Maxwell's equations for the time-varying case, which are then specialized to electrostatics and magnetostatics. These chapters will provide the student with an overall framework for what is to come and show him or her why electrostatics and magnetostatics are special cases of the more general time-varying case.

Chapter 6 deals with time-varying fields and sets the stage for the material in Chapters 7 through 9. Chapter 7 covers plane-wave propagation in dielectric and conducting media, and Chapter 8 covers reflection and transmission at discontinuous boundaries and introduces the student to fiber optics, waveguides, and resonators. In Chapter 9, the student is introduced to the principles of radiation by currents flowing in wires, such as dipoles, as well as to radiation by apertures, such as a horn antenna or an opening in an opaque screen illuminated by a light source.

To give the student a taste of the wide-ranging applications of electromagnetics in today's technological society, Chapter 10 concludes this book with presentations of two system examples: satellite communication systems and radar sensors.

The material in this book was written for a two-semester sequence of six credits, but it is possible to trim it down to generate a syllabus for a one-semester, four-credit course. The accompanying table provides syllabi for each of these options.

Suggested Syllabi

		Two-Semester Syllabus 6 credits (42 contact hours per semester)		One-Semester Syllabus 4 credits (56 contact hours)	
	Chapter	Sections	Hours	Sections	Hours
1	Introduction: Waves and Phasors	All	4	All	4
2	Transmission Lines	All	12	2-1 to 2-8 and 2-11	8
3	Vector Analysis	All	8	All	8
4	Electrostatics	All	8	4-1 to 4-10	6
5	Magnetostatics	All	7	5-1 to 5-5 and 5-7 to 5-8	5
	Exams		3		2
		Total for first semester	42		
6	Maxwell's Equations for Time-Varying Fields	All	6	6-1 to 6-3, and 6-6	3
7	Plane-Wave Propagation	All	7	7-1 to 7-4, and 7-6	6
8	Wave Reflection and Transmission	All	9	8-1 to 8-3, and 8-6	7
9	Radiation and Antennas	All	10	9-1 to 9-6	6
10	Satellite Communication Systems and Radar Sensors	All	5	None	—
	Exams		3		1
		Total for second semester	40	Total	56
	Extra Hours		2		0

内容安排与建议教学进度

本书首先在电气工程专业三年级学生应已熟悉的知识与电磁学内容之间建立联系。通常，学生在选修电磁学课程之前，已经学习过一门或多门电路课程，因此应熟悉电路分析、欧姆定律、基尔霍夫电流定律和电压定律等内容。

传输线是连接电路与电磁学的自然桥梁。学生暂时不必处理矢量或场，就可以运用熟悉的概念，学习波的运动、功率的反射与传输、相量、阻抗匹配，以及导波结构中的许多传播特性。这些概念将帮助学生理解第 7 至第 9 章中平面波在自由空间和材料介质中的传播。传输线安排在第 2 章，第 1 章则先复习复数与相量分析。

第 3 至第 5 章介绍矢量分析、静电学与静磁学。静电学一章从时变情况下的麦克斯韦方程组出发，再将其简化到静电场和静磁场。这样既建立了后续内容的总体框架，也说明静电学和静磁学是一般时变情况的特例。

第 6 章讨论时变场，为第 7 至第 9 章奠定基础。第 7 章介绍平面波在电介质和导电介质中的传播；第 8 章介绍波在不连续边界上的反射与透射，以及光纤、波导和谐振腔；第 9 章介绍导线电流的辐射，例如偶极子的辐射，以及口径辐射，例如喇叭天线或受光源照射的不透明屏上开口的辐射。

第 10 章通过卫星通信系统和雷达传感器两个系统实例，展示电磁学在当代技术社会中的广泛应用。

本书原按两学期、共 6 学分的课程编写，也可以删减为一学期、4 学分的课程。下面分别列出这两种课程的建议教学安排。

建议教学安排

章节	两学期：6 学分		一学期：4 学分	
	范围	学时	范围	学时
第1章 绪论：波与相量	全部	4	全部	4
第2章 传输线	全部	12	2-1 至 2-8；2-11	8
第3章 矢量分析	全部	8	全部	8
第4章 静电学	全部	8	4-1 至 4-10	6
第5章 静磁学	全部	7	5-1 至 5-5；5-7 至 5-8	5
考试		3		2
第一学期合计		42		
第6章 时变场的麦克斯韦方程组	全部	6	6-1 至 6-3；6-6	3
第7章 平面波传播	全部	7	7-1 至 7-4；7-6	6
第8章 波的反射与透射	全部	9	8-1 至 8-3；8-6	7
第9章 辐射与天线	全部	10	9-1 至 9-6	6
第10章 卫星通信系统与雷达传感器	全部	5	不安排	-
考试		3		1
第二学期合计 / 单学期合计		40		56
机动学时		2		0

两学期方案每学期授课 42 学时；一学期方案共 56 学时。表中范围编号沿用原书。